

Aurélien GAILLARD BTS SIO SISR ALT

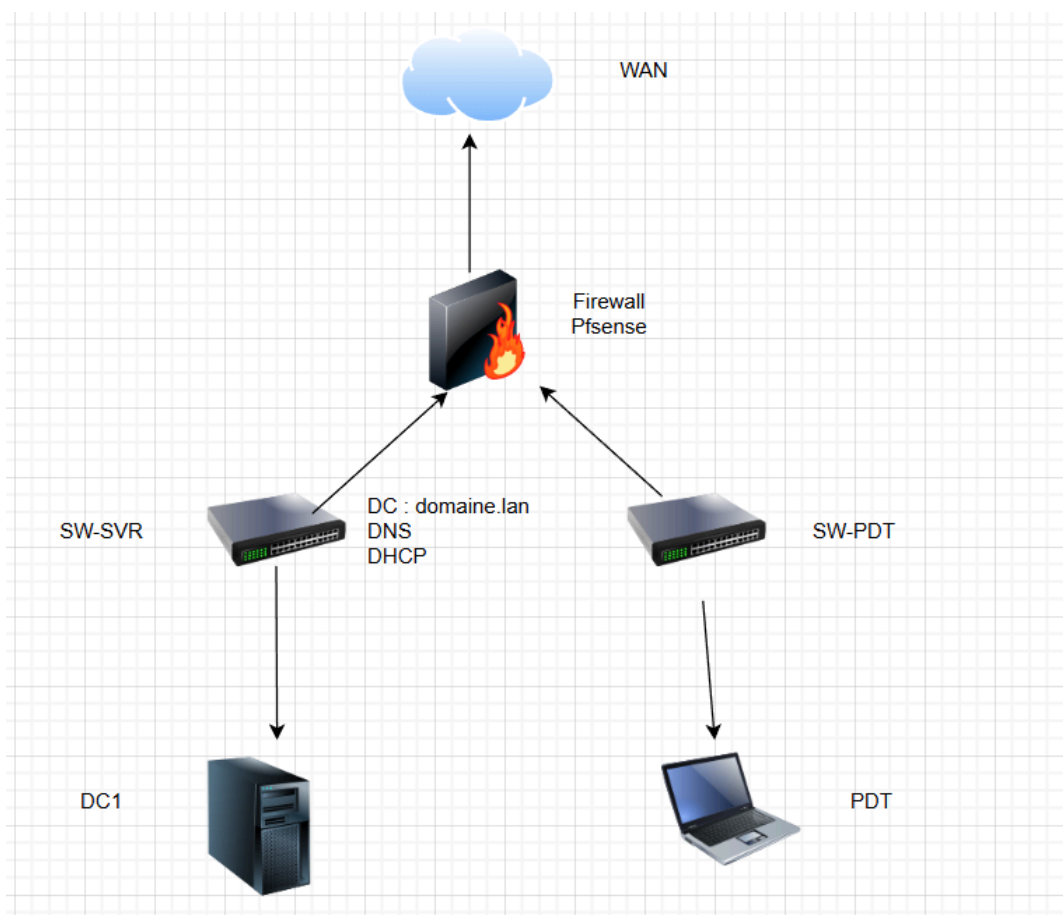
Procédure : Mise en place d'un pare-feu PFSENSE

L'objectif est d'installer un pare-feu PFSENSE, de paramétrer les interfaces pour les serveurs ainsi que les postes et également de créer des règles de filtrage.

1ère Étape : La création du plan d'adressage de l'architecture :

Équipement	Interface	Dernière IP	Masque	Passerelle (Gateway)
Firewall	eth0	WAN	WAN1	FAI
	eth1	192.168.2.30	255.255.255.224	FAI
	eth2	192.168.1.254	255.255.255.0	FAI
DC1	NIC	192.168.2.2	255.255.255.224	192.168.2.30
PDT	NIC	DHCP (192.168.1.0)	255.255.255.0	192.168.1.254

Schéma représentatif de notre architecture réseau :



Configuration des 2 interfaces LAN et WAN sur l'interface :

LAN SVR : 192.168.2.30 /27

LAN PDT : 192.168.1.254 /24

WAN : 192.168.240.20 /24

```
Press ENTER to continue.
UMware Virtual Machine - Netgate Device ID: 61bf83df007cd35584d7
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on FW-Site1 ***

WAN (wan)      -> em0      -> v4: 192.168.240.20/24
LANSVR (lan)   -> em1      -> v4: 192.168.2.30/27
LANPDT (opt1)  -> em2      -> v4: 192.168.1.254/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

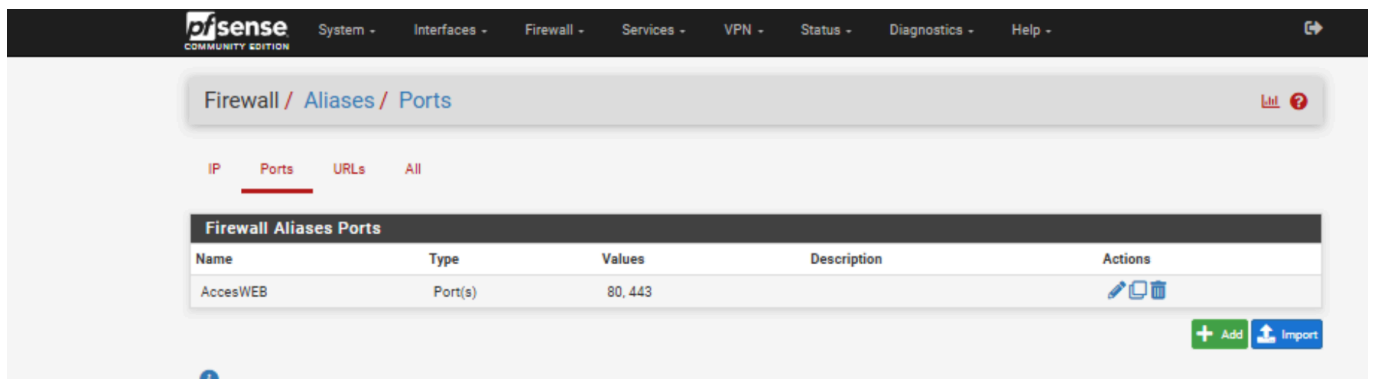
Enter an option:
Message from syslogd@FW-Site1 at Nov 27 13:50:37 ...
php-fpm[3971]: /index.php: Successful login for user 'admin' from: 192.168.2.2 (Local Database)
```

Aperçu des interfaces sur l'interface graphique de PFSense :

Interfaces  			
 WAN		1000baseT <full-duplex>	192.168.240.20
 LANSVR		1000baseT <full-duplex>	192.168.2.30
 LANPDT		1000baseT <full-duplex>	192.168.1.254

Création de la règle pour le trafic WEB

Tout d'abord nous allons procéder à la création d'un Alias afin de réunir les ports 80 et 443 en une seule entité :



The screenshot shows the pfSense Firewall Aliases Edit page. The breadcrumb trail is "Firewall / Aliases / Edit". The "Properties" section contains the following fields:

- Name:** AccesWEB. Below the field, it says: "The name of the alias may only consist of the characters 'a-z, A-Z, 0-9 and _'."
- Description:** (empty field). Below the field, it says: "A description may be entered here for administrative reference (not parsed)."
- Type:** Port(s) (selected from a dropdown menu).

The "Port(s)" section contains a hint: "Enter ports as desired, with a single port or port range per entry. Port ranges can be expressed by separating with a colon." Below the hint, there are two rows of input fields:

Port	Entry added Thu, 27 Nov 2025 15:57:30 +0100	Delete
80	Entry added Thu, 27 Nov 2025 15:57:30 +0100	
443	Entry added Thu, 27 Nov 2025 15:57:30 +0100	

At the bottom, there are buttons for "Save", "Export to file", and "+ Add Port".

Création de la règle :

Firewall / Rules / Edit

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LANPDT

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP/UDP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LANPDT subnets

Source Address

/

Display Advanced

The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

Any

Destination Address

/

Destination Port Range

(other)

AccesWEB

(other)

AccesWEB

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Destination

Destination

☐ Invert match

Any

Destination Address
/

Destination Port Range

(other)

AccesWEB

(other)

AccesWEB

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

1764252688

Created

11/27/25 15:11:28 by admin@192.168.2.2 (Local Database)

Updated

11/27/25 15:59:02 by admin@192.168.2.2 (Local Database)

Configuration des règles DNS dans le Firewall :

Création d'un Alias "DNSlocal" :

Firewall / Aliases / IP

IP
Ports
URLs
All

Firewall Aliases IP

Name	Type	Values	Description	Actions
DNSlocal	Host(s)	192.168.2.2		

Add
Import

Pour autoriser l'accès au DNS, il faut créer la même règle pour chaque interface :

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LANPDT

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

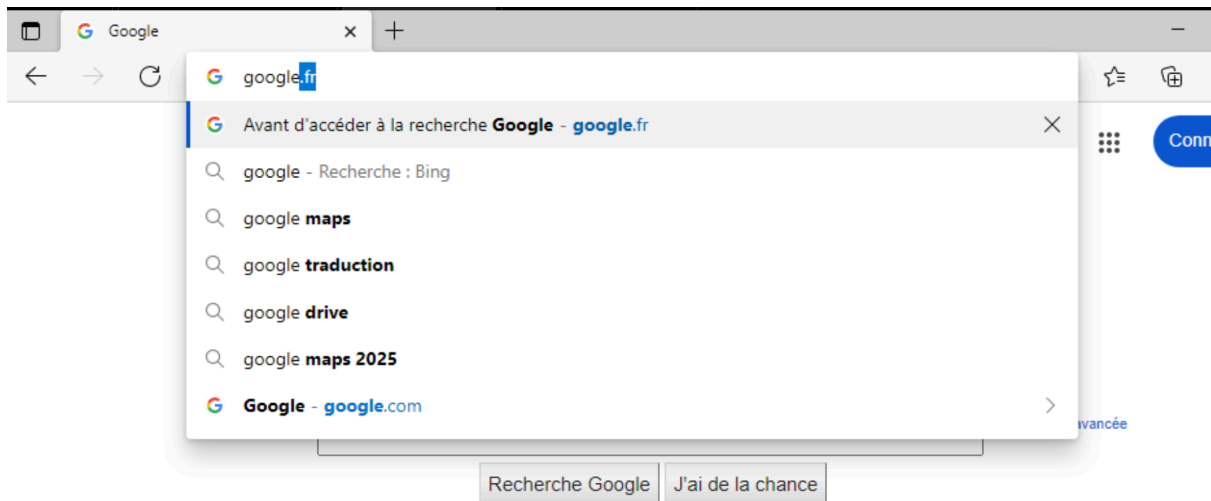
Select the Internet Protocol version this rule applies to.

Protocol

UDP

Choose which IP protocol this rule should match.

Test fonctionnement DNS :



```
C:\Users\admin>nslookup
Serveur par défaut : UnKnown
Address: 192.168.2.2
```

Activation et configuration d'un relay DHCP :

Services / DHCP Relay

DHCP Relay Configuration

Enable ☒ Enable DHCP Relay

Downstream Interfaces

WAN
LANSVR
LANPDT

Interfaces without an IPv4 address will not be shown.

CARP Status VIP

none

DHCP Relay will be stopped when the chosen VIP is in BACKUP status, and started in MASTER status.

☐ Append circuit ID and agent ID to requests
Append the circuit ID (interface number) and the agent ID to the DHCP request.

Upstream Servers

192.168.2.2

+ Add Upstream Server

The IPv4 addresses of the servers to which DHCP requests are relayed.

Save

Plage adresse DHCP sur le serveur AD :

DHCP

Fichier Action Affichage ?

	Adresse IP de début	Adresse IP de fin	Description
DC1.ief2i.lan	192.168.1.1	192.168.1.254	Plage d'adresses pour la distribution
IPv4	192.168.1.254	192.168.1.254	Adresses IP exclues de la distribution
Étendu			
Poc			
Bau			
Rés			
Op			
Str			
Option			
Stratég			
Filtres			
IPv6			

Création d'une règle pour autoriser le trafic afin d'ajouter un pc au domaine :

Tout d'abord, crée l'alias en ajoutant les ports nécessaires à l'ajout d'un pc sur un domaine :

The screenshot shows the 'Firewall / Aliases / Edit' configuration page. The 'Properties' section includes fields for 'Name' (AjoutDomaine), 'Description', and 'Type' (Port(s)). Below this is the 'Port(s)' section, which contains a table of ports. A hint at the top of the table states: 'Enter ports as desired, with a single port or port range per entry. Port ranges can be expressed by separating with a colon.'

Port	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
135	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
389	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
636	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
3268	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
3269	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
53	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
88	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
445	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete
1024-65535	Entry added Thu, 11 Dec 2025 12:09:41 +0100	Delete

At the bottom of the interface are three buttons: 'Save', 'Export to file', and 'Add Port'.

Création de la règle en utilisant l'alias :

Firewall / Rules / Edit

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LANPDT

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP/UDP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LANPDT subnets

Source Address

/

Display Advanced

The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

LANSVR subnets

Destination Address

/

Destination Port Range

(other)

AjoutDomaine

(other)

AjoutDomaine

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Destination

Destination

☐ Invert match

LANSVR subnets

Destination Address

/

Destination Port Range

(other)

AjoutDomaine

(other)

AjoutDomaine

From

Custom

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the Status: System Logs: Settings page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

1765451550

Created

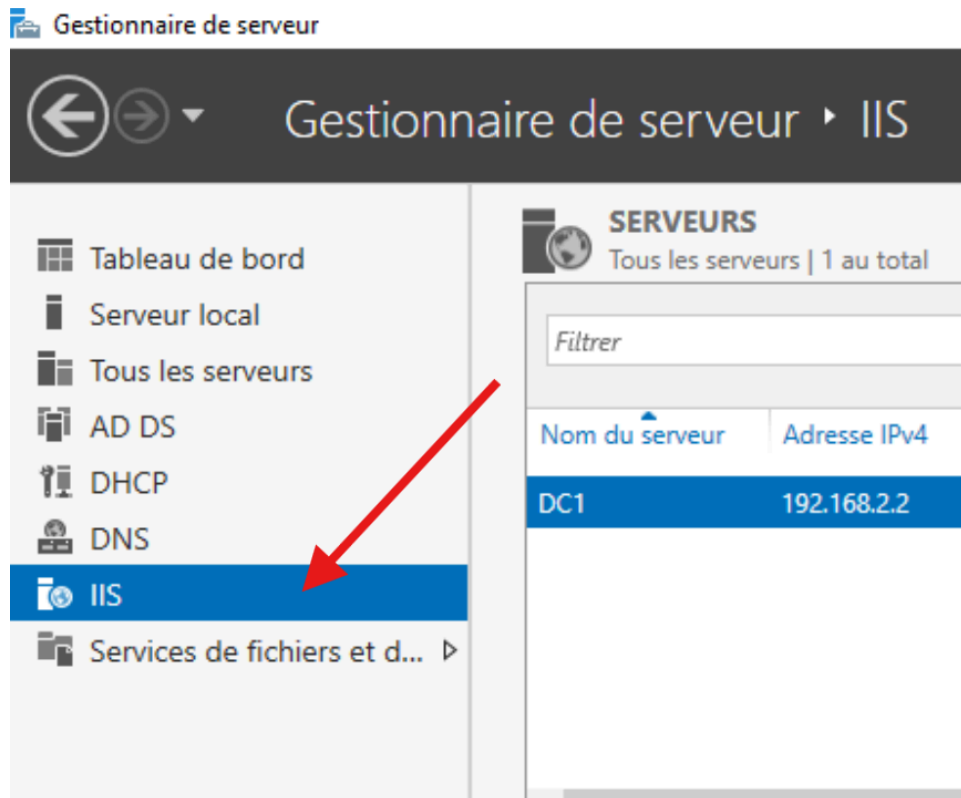
12/11/25 12:12:30 by admin@192.168.2.2 (Local Database)

Updated

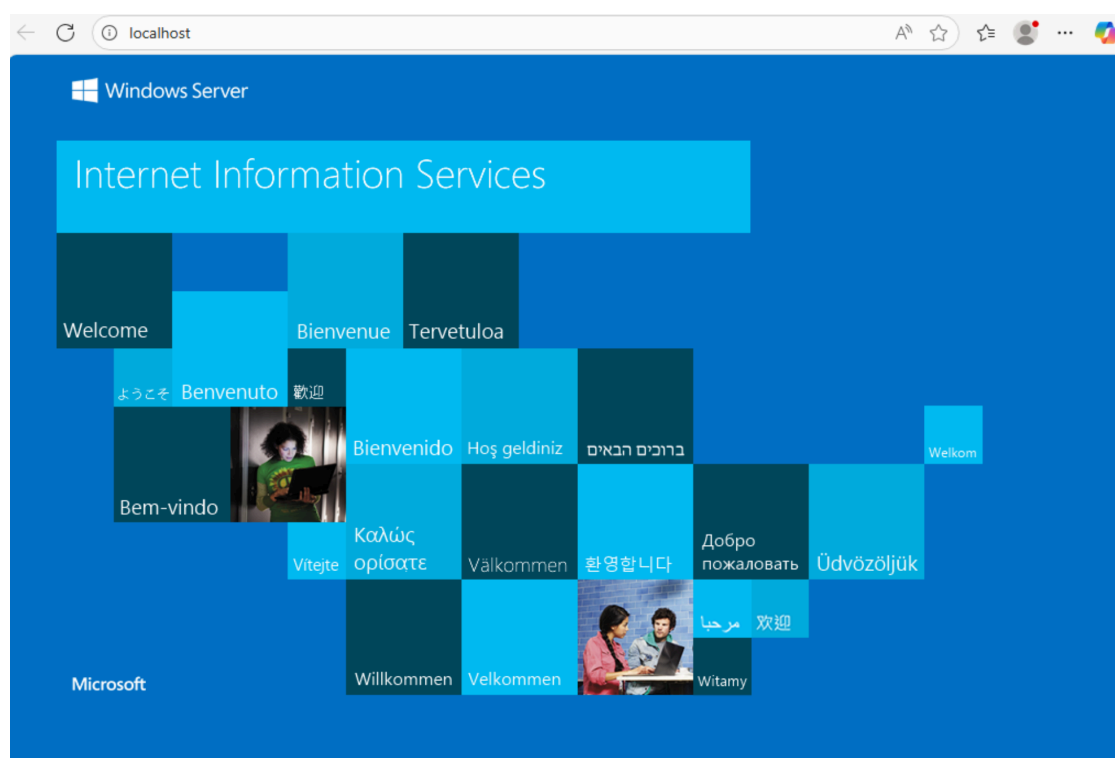
12/11/25 12:24:45 by admin@192.168.2.2 (Local Database)

Mise en place d'une règle horaire permettant l'accès au serveur Web interne

Tout d'abord, il faut installer le service IIS sur le serveur



Une fois l'installation, taper l'adresse suivant pour tester : <http://localhost>



Création de la plage horaire sur Pfsense :

Schedule Name	Acces_site_web																																																				
The name of the schedule may only consist of the characters "a-z, A-Z, 0-9 and _".																																																					
Description																																																					
A description may be entered here for administrative reference (not parsed).																																																					
Month	December_25																																																				
Date	<table><thead><tr><th colspan="7">December_2025</th></tr><tr><th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th><th>Sat</th><th>Sun</th></tr></thead><tbody><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr><tr><td>15</td><td>16</td><td>17</td><td>18</td><td>19</td><td>20</td><td>21</td></tr><tr><td>22</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td></tr><tr><td>29</td><td>30</td><td>31</td><td></td><td></td><td></td><td></td></tr></tbody></table>				December_2025							Mon	Tue	Wed	Thu	Fri	Sat	Sun	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31				
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22	23	24	25	26	27	28																																															
29	30	31																																																			
Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of that weekday.																																																					
Time	0	00	23	59																																																	
	Start Hrs	Start Mins	Stop Hrs	Stop Mins																																																	
Select the time range for the day(s) selected on the Month(s) above. A full day is 0:00-23:59.																																																					
Time range description																																																					
A description may be entered here for administrative reference (not parsed).																																																					
+ Add Time		Clear selection																																																			
Configured Ranges																																																					
Mon - Fri	8:00	18:00		Delete																																																	
Day(s)	Start time	Stop time	Description																																																		

Création de la règle d'accès à la page web de 8h à 18h :

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LANPDT

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LANPDT subnets

Source Address

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.

Destination

Destination

☐ Invert match

Address or Alias

Serveur_web

/

Destination Port Range

(other)

From

Custom

(other)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Schedule

Acces_site_web

Leave as 'none' to leave the rule enabled all the time.

Il faut également créer une autre règle qui bloque l'accès au serveur web sans préciser d'heure cette fois ci :

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LANPDT

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LANPDT subnets

Source Address

/

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.

Advanced Options

Display Advanced

Rule Information

Tracking ID

1765462507

Created

12/11/25 15:15:07 by admin@192.168.2.2 (Local Database)

Updated

12/11/25 15:15:07 by admin@192.168.2.2 (Local Database)

Save

Ordre des règles :

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 0/212 KiB	IPv4 TCP	LANPDT subnets	*	Serveur_web	AccesWEB	*	none	Acces_site_web		    
☐	✗ 0/3 KiB	IPv4 TCP	LANPDT subnets	*	Serveur_web	*	*	none			    
☐	✓ 2/2.10 GiB	IPv4 TCP/UDP	LANPDT subnets	*	*	AccesWEB	*	none			    
☐	✓ 0/659 KiB	IPv4 TCP/UDP	LANPDT subnets	*	LANSVR subnets	AjoutDomaine	*	none			    
☐	✓ 0/180 KiB	IPv4 UDP	LANPDT subnets	*	DNSlocal	53 (DNS)	*	none			    

